

Program ASASipintar Sesi 2026/2027

Pusat PERMATA@pintar Negara Universiti Kebangsaan Malaysia

TUTORIAL 1: WEEK 1 Biochemistr

Name:

Date:

Set:

1. Discuss:

- a) What is electronegativity?
- b) Why electronegativity is important in covalent bonds?
- c) Explain these statements below, with appropriate examples.
 - i. Polar covalent bond
 - ii. Nonpolar covalent bond
 - iii. Why water is considered as polar molecules?
 - iv) Ionic Bonds
 - v) Hydrogen Bonds
 - vi) Van Der Waals interactions

2. Water is important for all living organisms.

- a) Describe how the properties of water contribute to the following:
 - i. transpirations in plants
 - ii. thermoregulation in endotherms animals
 - iii. for animals/fish living in lakes in winter.
- b) Describe the High specific heat feature of water. Why is it so important to us human?

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